

TalentDNA AI

Intelligent Candidate Discovery Engine

Team EduBotX

Problem Statement

Traditional recruitment systems rely heavily on keyword matching, causing strong candidates to be overlooked while resume-stuffing profiles receive inflated rankings. Recruiters also lack visibility into candidate engagement, profile quality, and hiring readiness.

TalentDNA AI addresses these limitations through semantic retrieval, behavioral signals, evidence-based ranking, and explainable recommendations.

Solution Overview

TalentDNA AI ranks candidates using a multi-stage pipeline:

Candidate Dataset



Candidate Parser



Career Intelligence



Embedding Engine



FAISS Retrieval



Role Intelligence



Evidence Matching



Behavioral Signals



Integrity Validation



Hybrid Ranking



Explainability Layer



Top 100 Candidates

Core Components

Semantic Retrieval

Candidate profiles are converted into vector embeddings and indexed using FAISS. The job description is embedded and used to retrieve the most relevant candidates from a pool of 100,000 profiles.

Career Intelligence

Extracts experience, career progression, role history, and professional background signals to better understand candidate suitability.

Evidence Matching

Identifies technical evidence from candidate profiles including retrieval systems, machine learning, ranking systems, vector databases, and production AI experience.

Behavioral Signal Engine

Incorporates Redrob behavioral signals such as recruiter engagement, response rates, profile completeness, activity levels, interview completion rates, and hiring readiness.

Integrity Validation

Applies profile quality checks and scoring adjustments based on profile consistency and candidate signals.

Explainability Layer

Generates human-readable reasoning for every ranked recommendation to improve recruiter trust and transparency.

Ranking Methodology

Final ranking combines multiple signals:

Semantic Relevance

+ Role Alignment

+ Evidence Strength

+ Behavioral Signals

+ Integrity Score

The system produces a deterministic final score used to rank candidates from highest to lowest suitability.

Benchmark Results

100,000 Candidates Indexed

1,000 Candidates Retrieved via FAISS

Top 100 Ranked Candidates Generated

CPU-Only Inference

No External APIs During Ranking

Deterministic Output

Explainable Recommendations

Compliance

- ✓ Runtime within hackathon limits
 - ✓ CPU-only execution
 - ✓ No hosted LLM/API calls during ranking
 - ✓ Reproducible ranking pipeline
 - ✓ Explainable candidate recommendations
 - ✓ Valid Top-100 submission format
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Team EduBotX

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